

ECE 4644 / 5610 Satellite Communications

Homework Assignment #4 SOLUTIONS

Note: For all problems take the radius of Earth, R_E , to be 6378.137 km.

Problem #1

A LEO satellite completes a circular orbit in 95 min and is in darkness for half of each orbit. The satellite requires 1.5 kW of electrical power, which must be supplied by the batteries when the satellite is in darkness. When the satellite is in sunlight, its solar cells must supply sufficient power to run the satellite and also to charge the batteries. Charging the batteries requires 1.25 times the power that is needed to run the satellite.

- How much power must the solar cells provide while the satellite is in sunlight?
- The solar cells supply current at 50 V. What is the maximum current that they deliver?
- The satellite is in darkness for precisely half of each orbit. What minimum capacity must the batteries have, in ampere-hours? Make a recommendation.
- Given that the incident solar power is 1.36 kW/m^2 , the efficiency of the solar cells is 30%, and the cells are always oriented to maintain normal incidence, what area of solar cells is required?

Problem #2

In a type-written report meant for the company CEO that totals no more than one page with 12-point font and single-spaced, describe the major subsystems of a communications satellite. Your description of each subsystem should describe its function, instrumentation, and any special technical issues or choices. Be quantitative where possible. Provide a brief introduction to the report (what is a communications satellite?) and conclude with a statement about what limits the operational life of a communications satellite. Be sure to identify yourself and the project. The format of your bio sketch (HW#0) might be applicable here. (IMPORTANT: write in your own words using the concepts and terms developed in class.)

Problem #3

In class we discussed the processing of signal for a 6/4 GHz satellite link through a single conversion system using *low side injection*, $f_{\text{up}} = 6.000 \text{ GHz}$, $f_{\text{LO}} = 2.225 \text{ GHz}$, $f_{\text{down}} = 3.775 \text{ GHz}$. We now modify the design to a double conversion system with an IF frequency of 1.000 GHz using *high side injection* and the same uplink and downlink frequencies. The bandwidth available to the satellite is still 500 MHz.

- Determine the values of the LO frequencies needed for the two stages of mixing.
- Analyze the situation for the effect of an image frequency arriving at the first mixer, i.e., show that there is another frequency besides 6.000 GHz that translates to the IF frequency. Given that the bandwidth of the uplink is 500 MHz, describe a suitable image rejection filter and state where you would place it.
- Explain why a filter is placed after each mixer. Describe suitable filters in terms of center frequency and bandwidth.

- d. Now assemble the components into a block diagram showing the processing of signal from uplink antenna to downlink antenna. Include amplifiers at suitable stages (do not specify gain). Label all components and indicate frequencies. (Hint: borrow from the lecture slides.)
- e. If the transponder channels are 27 MHz in bandwidth and separated in their center frequencies by 30 MHz, how many transponders can the satellite carry?
- f. Explain why 100 W is a more plausible figure for the output power of one of the transponders than 1 kW.
- g. Suppose the input power to the transponder from the receiving antenna is 10^{-11} W and the output power to the transmitting antenna is 100 W. What is the net gain through the transponder?
- h. Explain why the amplification at the last stage of the transponder is usually split between two amplifiers (often labeled LPA and HPA, explain the labels)

Problem #4

A communications link has been set up at a frequency of 15 GHz using a circular transmitting antenna with a diameter of 1.5 m, an efficiency $\eta_A = 60\%$, and a transmit power of 100 W.

- a. State the frequency band of the system.
- b. Solve for the gain of the antenna (exact result) and estimate its beamwidth.
- c. At the position of a satellite passing overhead in LEO orbit with $h = 600$ km, solve for the power density in the arriving signal.
- d. Determine the power that could be delivered by a receiving antenna at the satellite (in Watts and dBW) if it has an effective collecting area of 1.5 m^2 .

Problem #5

A communications system is designed to operate in the middle of C band. The transmit power delivered to the antenna is 1 kW. The antenna is rectangular in shape with a horizontal dimension of 1 m and a vertical dimension of 0.5 m. The receiving antenna is identical and located at a distance of 750 km.

- a. Estimate the beamwidths of the antenna in the horizontal and vertical planes and its gain.
- b. Find the EIRP of the transmitter in dBW and watts.
- c. Determine the path loss in dB.
- d. Assuming no losses due to the antennas or to atmospheric effects, find the power available from the receiving antenna.
- e. Suppose each antenna has a loss of 1 dB and atmospheric attenuation totals 0.2 dB, now find the power available from the receiving antenna.
- f. Suppose we cannot avoid these losses but still require the receiver power that was found in part (d) for reliable operation. What is the maximum distance that is possible?

PROBLEM 1

A LEO satellite completes a circular orbit in 95 min and is in darkness for half of each orbit. The satellite requires 1.5 kW of electrical power, which must be supplied by the batteries when the satellite is in darkness. When the satellite is in sunlight, its solar cells must supply sufficient power to run the satellite and also to charge the batteries. Charging the batteries requires 1.25 times the power that is needed to run the satellite.

a. How much power must the solar cells provide while the satellite is in sunlight?

The power necessary to run the satellite is given as 1.5 kW. The batteries require 1.25 the power necessary to run the satellite to charge, or 1.875 kW. Therefore, the solar panels must produce the 1.5 kW necessary to power the satellite for that portion of the orbit in addition to the 1.875 kW necessary to charge. In total, **3.375 kW** of power must be generated by the solar panels while in sunlight.

b. The solar cells supply current at 50 V. What is the maximum current that they deliver?

The following equation is used to calculate the current required to supply 3.375 kW (the maximum power) at 50 V:

$$I_{max} = \frac{P}{V} = \frac{3375 \text{ W}}{50 \text{ V}} = \mathbf{67.5 \text{ A}}$$

c. The satellite is in darkness for precisely half of each orbit. What minimum capacity must the batteries have, in ampere-hours? Make a recommendation.

The orbital period of the satellite is 95 minutes, so it is in eclipse for 47.5 minutes each orbit. This is also equal to 0.792 hours. To supply 1.5 kW (30 A) for 0.792 hours, 23.75 AH are needed during eclipse if the batteries are discharged to zero every revolution. Using a typical value of 70% discharge since the batteries cannot be fully drained, a total capacity of **33.93 AH** would be required. This is the recommended amount to ensure margin.

d. Given that the incident solar power is 1.36 kW/m², the efficiency of the solar cells is 30%, and the cells are always oriented to maintain normal incidence, what area of solar cells is required?

For the given incident solar flux, the total area of the solar panels required can be calculated as shown, where the numerator is the required power and the denominator is the product of the solar cell efficiency and the incident solar flux:

$$A = \frac{3.375 \text{ kW}}{0.3 \times 1.36 \text{ kW/m}^2} = \mathbf{8.272 \text{ m}^2}$$

PROBLEM 2

In a type-written report meant for the company CEO that totals no more than one page with 12-point font and single-spaced, describe the major subsystems of a communications satellite. Your description of each subsystem should describe its function, instrumentation, and any special technical issues or choices. Be quantitative where possible. Provide a brief introduction to the report (what is a communications satellite?) and conclude with a statement about what limits the operational life of a communications satellite. Be sure to identify yourself and the project. The format of your bio sketch (HW#0) might be applicable here. (IMPORTANT: write in your own words using the concepts and terms developed in class.)

The essay for this assignment is attached at the end of the submission.

PROBLEM 3

In class we discussed the processing of signal for a 6/4 GHz satellite link through a single conversion system using low side injection, $f_{up} = 6.000$ GHz, $f_{LO} = 2.225$ GHz, $f_{down} = 3.775$ GHz. We now modify the design to a double conversion system with an IF frequency of 1.000 GHz using high side injection and the same uplink and downlink frequencies. The bandwidth available to the satellite is still 500 MHz.

a. Determine the values of the LO frequencies needed for the two stages of mixing.

For high side injection, the LO frequencies can be calculated using the signal and IF frequencies:

$$f_{LO} = f_{signal} + f_{IF}$$

First, the LO frequency for downconversion is calculated:

$$f_{LO1} = f_{up} + f_{IF} = 6.000 + 1.000 = 7.000 \text{ GHz}$$

Next, the second LO frequency for upconversion is calculated:

$$f_{LO2} = f_{down} + f_{IF} = 3.775 + 1.000 = 4.775 \text{ GHz}$$

b. Analyze the situation for the effect of an image frequency arriving at the first mixer, i.e., show that there is another frequency besides 6.000 GHz that translates to the IF frequency. Given that the bandwidth of the uplink is 500 MHz, describe a suitable image rejection filter and state where you would place it.

An image frequency is $2 \times IF$ away from the signal frequency. For the first mixer, the signal frequency is $f_{up} = 6.000$ GHz. For high side injection, the image signal will be above the signal frequency:

$$f_{IM} = f_{up} + (2 \times f_{IF}) = 6.000 + 2.000 = 8.000 \text{ GHz}$$

The difference between this image frequency and the first LO frequency is equal to the IF frequency, demonstrating that it would potentially interfere. The image rejection filter must be placed before the first mixer so that the image frequency does not pass through the mixer and become part of the same intermediate frequency. A bandpass filter could be used as the image rejection filter. For example, a bandpass filter centered at 6.0 GHz with a bandwidth of 500 MHz would allow frequencies between 5.75 and 6.25 only, suitably rejecting f_{IM} .

c. Explain why a filter is placed after each mixer. Describe suitable filters in terms of center frequency and bandwidth.

Filters are placed after each mixer to select for the desired frequency given that the mixer will produce a sum and difference frequency. To select for only the frequency that is desired for each stage, a filter must be applied. After the first mixer, the only desired frequency is $f_{IF} = 1$ GHz. A lowpass or bandpass filter could be used to select for this frequency. For example a bandpass filter centered at 1 GHz with a bandwidth of 500 MHz would only allow signals from 0.75 to 1.25 GHz. Similarly, after the second mixer, the downlink frequency of 3.775 GHz would be required. A similar bandpass filter centered at 3.775 GHz with a bandwidth of 500 MHz would successfully select for the desired downlink frequency. In summary:

- Filter 1: Center Frequency 1 GHz, Bandwidth 500 MHz
- Filter 2: Center Frequency 3.775 GHz, Bandwidth 500 MHz

d. Now assemble the components into a block diagram showing the processing of signal from uplink antenna to downlink antenna. Include amplifiers at suitable stages (do not specify gain). Label all components and indicate frequencies. (Hint: borrow from the lecture slides.)

The format from the lecture slides was borrowed and labeled to detail the double conversion system described in this problem, as shown in Figure 1.

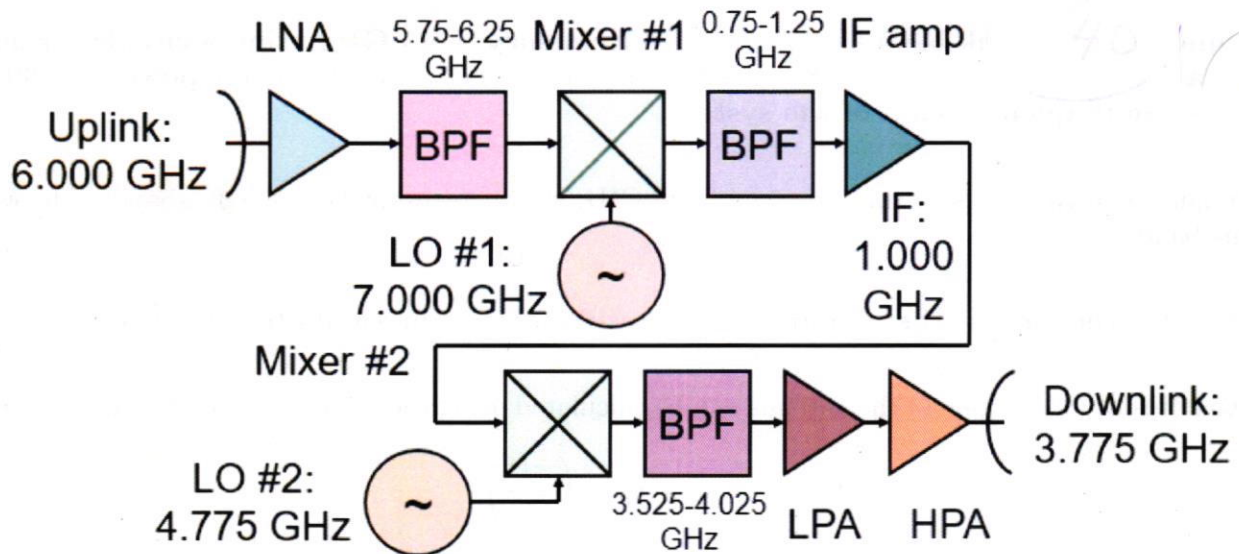


FIGURE 1. Double Conversion System with High Side Injection

e. If the transponder channels are 27 MHz in bandwidth and separated in their center frequencies by 30 MHz, how many transponders can the satellite carry?

For a bandwidth of 500 MHz with 27 MHz channels and 30 MHz separation, 16 channels can fit within the given bandwidth. Therefore, the satellite can carry **16 transponders**.

f. Explain why 100 W is a more plausible figure for the output power of one of the transponders than 1 kW.

If the transponders each had an output power of 1 kW, then 16 kW would be required in total just to power the number of transponders determined in this problem. This would be on the high end for the total amount of power cited in the lecture as generated by typical GEO satellite solar panels. Since other satellite systems must also be powered, it does not make sense for the transponders to take up most if not all of the power generated by the solar panels.

g. Suppose the input power to the transponder from the receiving antenna is 10^{-11} W and the output power to the transmitting antenna is 100 W. What is the net gain through the transponder?

Knowing the received power and the transmit power, the gain of the transponder can be calculated. This is easiest in dB units:

$$P_t = 100 \text{ W} = 20 \text{ dBW}, P_r = 10^{-11} \text{ W} = -110 \text{ dBW}$$

$$G = P_t - P_r = 20 - (-110) = \mathbf{130 \text{ dB}}$$

h. Explain why the amplification at the last stage of the transponder is usually split between two amplifiers (often labeled LPA and HPA, explain the labels)

These labels stand for Low Power Amplifier (LPA) and High Power Amplifier (HPA), respectively. As mentioned in the lecture notes, it is not advisable to have a single amplifier with more than 60 dB of gain as this will introduce oscillations from the amplifier. Instead, spreading the gain across an LPA and HPA can produce the desired gain without encountering this issue.

PROBLEM 4

A communications link has been set up at a frequency of 15 GHz using a circular transmitting antenna with a diameter of 1.5 m, an efficiency $\eta_A = 60\%$, and a transmit power of 100 W.

a. State the frequency band of the system.

The frequency of 15 GHz is within the range 12-18 GHz, which corresponds to the Ku-band. The wavelength for this band is ≈ 2 cm.

b. Solve for the gain of the antenna (exact result) and estimate its beamwidth.

The exact result for the gain of the antenna can be calculated using the following equation and the wavelength:

$$G = \eta_A \times \left(\frac{\pi D}{\lambda} \right)^2$$

$$f \times \lambda = 30 \rightarrow \lambda = \frac{30}{15} = 2$$

$$G = 0.6 \times \left(\frac{1.5\pi}{0.02} \right)^2 = 33310 = 10 \log_{10}(33310) = 45.22575 \text{ dB}$$

The beamwidth is estimated as shown, as is the gain (though the exact result was required as given above):

$$\theta_{3dB} \approx \frac{75\lambda}{D} = \frac{75 \times 0.02}{1.5} = 1 \text{ deg}$$

$$G \approx \frac{33000}{(\theta_{3dB})^2} = 33000 = 10 \log_{10}(33000) = 45.18514 \text{ dB}$$

c. At the position of a satellite passing overhead in LEO orbit with $h = 600$ km, solve for the power density in the arriving signal.

For a directional antenna, the power density can be calculated using the equation below:

$$Q_i = \frac{P_t G_t}{4\pi R^2}$$

$$Q_i = \frac{100 \text{ W} \times 33310}{4\pi(600000 \text{ m})^2} = 7.36313 \times 10^{-7} \text{ W/m}^2$$

d. Determine the power that could be delivered by a receiving antenna at the satellite (in Watts and dBW) if it has an effective collecting area of 1.5 m^2 .

The power density and effective collecting area are necessary to calculate the power that could be delivered by a receiving antenna:

$$P_r = Q_i A_{eff}$$

< excess decimal places

$$P_r = 7.36313 \times 10^{-7}(1.5) = 1.10447 \times 10^{-6} \text{ W} = 10 \log_{10}(1.10447 \times 10^{-6}) = -59.56846 \text{ dBW}$$

PROBLEM 5

A communications system is designed to operate in the middle of C band. The transmit power delivered to the antenna is 1 kW. The antenna is rectangular in shape with a horizontal dimension of 1 m and a vertical dimension of 0.5 m. The receiving antenna is identical and located at a distance of 750 km.

a. Estimate the beamwidths of the antenna in the horizontal and vertical planes and its gain.

The wavelength in the middle of C band is 5 cm:

$$f \times \lambda = 30 \rightarrow \lambda = \frac{30}{6 \text{ GHz}} = 5 \text{ cm}$$

Using the wavelength and dimensions of the antenna, the horizontal and vertical beamwidths are estimated:

$$\theta_{3dB,H} \approx \frac{75\lambda}{D_H} = \frac{75(0.05)}{1} = 3.75 \text{ deg}$$

$$\theta_{3dB,V} \approx \frac{75\lambda}{D_V} = \frac{75(0.05)}{0.5} = 7.5 \text{ deg}$$

The gain can also be estimated given the beamwidths:

$$G \approx \frac{33000}{(\theta_H \times \theta_V)} = 1173.33 = 10 \log_{10}(1173) = 30.69421 \text{ dB}$$

b. Find the EIRP of the transmitter in dBW and watts.

The EIRP of the transmitter is the product of the gain and the transmit power in linear units, or the sum in dB units:

$$P_t = 1000 \text{ W} = 10 \log_{10}(1000) = 30 \text{ dBW}$$

$$EIRP = P_t G_t = 1000 \times 1173.33 = 1173333 \text{ W} = 1.17 \times 10^6 \text{ W (linear)} = 1.17 \text{ MW}$$

$$EIRP = P_t G_t = 30 + 30.69421 = 60.69421 \text{ dBW (dB units)}$$

c. Determine the path loss in dB.

The path loss can be calculated using the following equation in both linear and dB form, where R and λ are both in meters:

$$L_P = \left(\frac{4\pi R}{\lambda} \right)^2 \rightarrow L_P = 10 \log_{10}(4\pi)^2 + 10 \log_{10} R^2 - 10 \log_{10} \lambda^2$$

$$L_P = 10 \log_{10}(4\pi)^2 + 10 \log_{10}(750000)^2 - 10 \log_{10}(0.05)^2 = 165.50602 \text{ dB}$$

d. Assuming no losses due to the antennas or to atmospheric effects, find the power available from the receiving antenna.

The power available from the receiving antenna can be found using the link equation:

$$P_r = P_t + G_t + G_r - L_p$$

$$P_r = 30 + 30.69421 + 30.69421 - 165.50602 = -74.1176 \text{ dBW}$$

e. Suppose each antenna has a loss of 1 dB and atmospheric attenuation totals 0.2 dB, now find the power available from the receiving antenna.

The link equation can be modified to include losses and atmospheric attenuation:

$$P_r = P_t + G_t + G_r - L_p - L_{misc}$$

$$P_r = 30 + 30.69421 + 30.69421 - 165.50602 - (1 + 1 + 0.2) = -76.3176 \text{ dBW}$$

f. Suppose we cannot avoid these losses but still require the receiver power that was found in part (d) for reliable operation. What is the maximum distance that is possible?

To compensate for the losses, the path loss must decrease by 2.2 dB. The new path loss can be converted to linear units to more easily solve for the final value:

$$L_P = 10 \log_{10}(4\pi)^2 + 10 \log_{10} R^2 - 10 \log_{10}(0.05)^2 = 163.30602 \text{ dB} = 2.14 \times 10^{16}$$

$$\left(\frac{4\pi R}{\lambda}\right)^2 = 2.14093 \times 10^{16} \rightarrow R = \left(\frac{\lambda}{4\pi}\right) \sqrt{2.14093 \times 10^{16}} = 582185 \text{ m} = 582 \text{ km}$$

The maximum distance possible is about 18 km less than the original distance when the various loss sources are considered.

MEMO

TO: Chief Executive Officer
FROM: Amanda J. Carlson

DATE: March 3, 2026

SUBJECT: Purpose and Subsystems of Communication Satellites

The purpose of this project is to describe the function, major subsystems, and operational lifetime considerations for communications satellites. A communications satellite is a spacecraft with a primary payload focused on receiving and re-transmitting communications signals as a relay between ground stations or as a link to other satellites. Although this is the primary mission, numerous other subsystems are necessary to support the communications system. ✓

The function of the Attitude and Orbit Control System (AOCS) is to control the orbit and orientation. The modern attitude control subsystem consists of momentum wheels or other instruments that can apply a rotational force about all three axes. Traditionally, a "spinner" configuration was also used in which a portion of the spacecraft's body was rotated at a rate between 30 and 100 rpm as another method of stabilization. The orbit control system uses thrusters in each direction to apply forces necessary to correct the satellite's orbit as perturbations occur. This is necessary to counteract inclination changes of approximately 0.85 degrees per year and drift in the E-W direction for a GEO satellite, or other perturbations in the case of LEO. < explain

The Telemetry, Tracking, Command, and Monitoring (TTC&M) system is utilized to control the satellite from the ground station and provide necessary data about system health to operators. Sensors that report status data from a variety of components, a command system to process directions from the ground, and multiple antennas are included in the TTC&M system. An omnidirectional UHF antenna can be used to acquire an initial link to the system while tumbling, while a narrow beam antenna in C or Ku-band is generally used for normal operations. This system is often duplicated for redundancy, as the spacecraft is unusable without it. ✓

The satellite's power system consists of the components necessary to convert sunlight to electricity and store it for later use. The main components of this system are the solar panels and batteries. Solar panels are made up of solar cells, which generally have an efficiency ranging from 20 to 35 percent initially before normal degradation. Batteries store electrical power needed to continue satellite operations through the eclipse period before the solar panels are deployed, and the necessary battery capacity will be determined based on the power needs of the satellite.

The primary payload, the communications system, fulfills the purpose mentioned previously to relay signals. It consists of antenna and transponder subsystems. Transponders amplify low-power signals (often below -90 dBW) and re-transmit them at a different frequency, and a communications satellite will have multiple transponder channels. Transponders are composed of filters, mixers, local oscillators, and amplifiers in addition to the antennas. Common antenna types include wire, horn, reflector, and phased array antennas, which are chosen based on their coverage and antenna pattern. For example, a global beam using a horn antenna covers the visible surface of the Earth, whereas a reflector antenna can produce spot beams. < what is typical for production? 5-15 kW

Additional subsystems are also necessary, but this report summarizes major components. It is important to note that the operational lifetime of a communications satellite is limited by the lifespan of individual components and the fuel necessary for stationkeeping. The effects of component degradation can be minimized by using space-qualified hardware and introducing redundancy into the design. { what freqs are used? C, Ku, Ka bands

what is typical?

~15 years

< Need to differentiate early on

between GEO sats and LEO constellations